

Prepare: FNRP Ch. 3 (multigroup equations); review Lesson 12

Primary objective. Construct multigroup cross sections and solve neutron balances for an infinite, homogeneous system.

1 Replace assumed flux ratios

Lesson 12 supplied normalized shapes and *assumed* the flux in each range. Now determine those fluxes from coupled balances. Number groups from **high to low energy**:

$$E_0 > E_1 > \dots > E_G, \quad \phi_g = \int_{E_g}^{E_{g-1}} \phi(E) dE.$$

The complete domain has $E_0 = \infty$ and $E_G = 0$. For our three-group examples, use the following finite bounds (eV):

$$(E_0, E_1, E_2, E_3) = (10^7, 10^5, 1, 10^{-3}).$$

Thus $(\phi_1, \phi_2, \phi_3) = (\phi_F, \phi_I, \phi_T)$. Finite cutoffs require attention to transfers outside the retained range.

2 Preserve each group reaction rate

For $\phi_g > 0$, define

Group averages

$$\Sigma_{x,g} = \frac{\int_{E_g}^{E_{g-1}} \Sigma_x(E) \phi(E) dE}{\phi_g},$$

$$(\nu \Sigma_f)_g = \frac{\int_{E_g}^{E_{g-1}} \nu(E) \Sigma_f(E) \phi(E) dE}{\phi_g}.$$

Use a normalized within-group shape to prepare these averages; then solve for ϕ_g (Lesson 12). The group sources are

$$\chi_g = \int_{E_g}^{E_{g-1}} \chi(E) dE, \quad s_g = \int_{E_g}^{E_{g-1}} s_{\text{ext}}'''(E) dE.$$

3 Keep the scattering direction straight

Destination $g \leftarrow$ incident group g'

$$\Sigma_{s,g \leftarrow g'} = \frac{\int_{E_g}^{E_{g'-1}} \phi(E') \int_{E_g}^{E_{g-1}} \Sigma_s(E' \rightarrow E) dE dE'}{\phi_{g'}}.$$

The denominator is the **incident-group flux**. Multiplying by $\phi_{g'}$ recovers the rate of transfer into g .

Checkpoint

With this numbering, downscatter means g _____ g' . In a scattering matrix whose rows are destinations, are the forced zero (upscatter) entries above or below the diagonal?

4 One balance per destination group

Integrate the spectrum equation over $E_g \leq E \leq E_{g-1}$:

Infinite-medium multigroup balance

$$\Sigma_{t,g} \phi_g = \sum_{g'=1}^G \Sigma_{s,g \leftarrow g'} \phi_{g'} + \chi_g \sum_{g'=1}^G (\nu \Sigma_f)_{g'} \phi_{g'} + s_g, \quad g = 1, \dots, G.$$

The fission term sums production over *all incident groups*; χ_g then distributes the newborn neutrons to destination g . Move self-scattering to the left by defining **removal**:

$$\Sigma_{r,g} = \Sigma_{t,g} - \Sigma_{s,g \leftarrow g}.$$

With number conservation and complete energy coverage,

$$\Sigma_{r,g} = \Sigma_{a,g} + \sum_{h \neq g} \Sigma_{s,h \leftarrow g}.$$

Removal includes absorption *and* scattering out of a group; it is not generally the same as absorption.

5 Three-group water tank

Approximate the ^{252}Cf source as $(s_1, s_2, s_3) = (10, 0, 0)$ in $\text{cm}^{-3} \text{s}^{-1}$. Ignore multiplication, leakage, intergroup upscatter, and transfers outside the finite energy range.

Solve from fast to thermal

Write each group balance and isolate its flux:

$$\phi_1 = \frac{\Sigma_{s,3 \leftarrow 1} \phi_1 + \Sigma_{s,3 \leftarrow 2} \phi_2}{\Sigma_{r,3}}.$$

Unlike the Lesson 12 example, the flux ratios are now outputs.

Check the complete balance

Add the three equations. Under the stated assumptions, show that

$$\sum_{g=1}^3 \Sigma_{a,g} \phi_g = 10 \text{ cm}^{-3} \text{s}^{-1}.$$

Why should removal cross sections *not* replace absorption cross sections in this check?

6 Use the scattering kinematics

For elastic scattering isotropic in the CM system from a stationary target of mass ratio A , let $\alpha = [(A - 1)/(A + 1)]^2$. Then

$$p(E' \rightarrow E) = \begin{cases} 1/[(1 - \alpha)E'], & \alpha E' \leq E \leq E', \\ 0, & \text{otherwise.} \end{cases}$$

This kernel cannot upscatter; it is not a thermal free-gas kernel. The allowed outgoing interval and destination group overlap over

$$\Delta E_g(E') = \left[\min(E_{g-1}, E') - \max(E_g, \alpha E') \right]_+, \quad [z]_+ = \max(z, 0).$$

Thus the probability of landing in group g is

$$P_g(E') = \frac{\Delta E_g(E')}{(1 - \alpha)E'}.$$

Incident-flux-weighted transfer

$$\sigma_{s,g \leftarrow g'} = \frac{\int_{E_{g'}}^{E_{g'-1}} \sigma_s(E') P_g(E') \phi(E') dE'}{\int_{E_{g'}}^{E_{g'-1}} \phi(E') dE'}.$$

For a mixture, compute each nuclide's transfer with its own α_i , then form $\Sigma_{s,g \leftarrow g'} = \sum_i N_i \sigma_{s,g \leftarrow g'}^i$.

An empty overlap is zero

Why would directly integrating from the larger lower bound to the smaller upper bound give an unphysical result?

7 Hydrogen: scatter out of group 2

Use the same three-group bounds, ^1H ($\alpha = 0$), $\sigma_s(E) = \sigma_0 = 20$ b, and $\phi(E') \propto 1/E'$ within incident group 2. Write

$$L = \ln(E_1/E_2) = \ln(10^5).$$

Set up the two nonzero integrals

$$\sigma_{s,2 \leftarrow 2} = \frac{\sigma_0}{L} \int_{E_2}^{E_1} \frac{E' - E_2}{(E')^2} dE',$$

$$\sigma_{s,3 \leftarrow 2} = \frac{\sigma_0}{L} \int_{E_2}^{E_1} \frac{E_2 - E_3}{(E')^2} dE'.$$

Explain the numerators using the allowed outgoing energy intervals.

8 Evaluate the scattering column

The group-to-group cross sections are

$$\begin{aligned} \sigma_{s,1 \leftarrow 2} &= 0, \\ \sigma_{s,2 \leftarrow 2} &= \sigma_0 \frac{L + E_2/E_1 - 1}{L}, \\ \sigma_{s,3 \leftarrow 2} &= \sigma_0 \frac{(E_2 - E_3)(1/E_2 - 1/E_1)}{L}. \end{aligned}$$

Compute the values

Destination	$\sigma_{s,g \leftarrow 2}$ (b)
$g = 1$ (fast)	0
$g = 2$ (intermediate)	_____
$g = 3$ (thermal)	_____

Most collisions stay within the very broad intermediate group; about 8.68% enter the retained thermal interval. These are conditional collision fractions, *not* ratios of steady-state group fluxes.

Where did the missing cross section go?

With $E_3 = 10^{-3}$ eV, some hydrogen collisions produce $0 < E < E_3$. Their contribution is

$$\sigma_{s,<E_3 \leftarrow 2} = \sigma_0 \frac{E_3(1/E_2 - 1/E_1)}{L} = \text{_____ b.}$$

Hence

$$\sum_{g=1}^3 \sigma_{s,g \leftarrow 2} = \sigma_0 - \sigma_{s,<E_3 \leftarrow 2} \approx 19.99826 \text{ b.}$$

How would the column sum change if the last group extended to $E = 0$?

9 Check the model, not just the algebra

For a complete set of outgoing energies,

$$\sum_g \Sigma_{s,g \leftarrow g'} = \Sigma_{s,g'}, \quad R_x = \sum_g \Sigma_{x,g} \phi_g.$$

For truncated groups, keep any out-of-range transfer as an additional loss, or state the approximation used to close the range. Do not silently claim exact conservation after discarding that transfer.

Takeaway

Multigroup balance determines the *amount* of flux in each group. The within-group spectra used to prepare cross sections remain modeling inputs. Fission couples the groups again; Lesson 14 uses that coupling to construct the k -eigenvalue problem.